TX-OMICS REVISION

Seven gene sets.
One auditable curation.

A concise record of the publication rules, pathway selection, identity checks, and overlap that produced the stocker-ready inputs.

7
stocker CSVs

2
analysis families

4
primary sets

3
GO-derived sets

Two curation families—analyzed separately

Primary evidence sets and GO term-grouped sets use different provenance and separate overlap analyses.

FAMILY 1 · PRIMARY CURATION SETS

01

6 genes

Mechanistic

Publication core + screen ranks 1–6

02

20 genes

Homeostatic H×R

15 History+/Rebound− + 5 opposite

03

97 genes

CSW 4+

Wake frequency ≥4, threshold-scoped

04

4 genes

HLH regulators

Four bHLH factors named in paper

FAMILY 2 · GO TERM-GROUPED SETS

05

99 genes

Ribosomal

DAVID term hits; approved keywords

06

167 genes

Mitochondrial

DAVID mito/metabolism term hits

07

5 genes

Immune

DAVID immune / Toll / Imd term hits

RULE

Keep them separate

Report overlap within each family—never as one seven-set comparison.

Two curation tracks converge on one controlled folder

They share identity gates and file standards—but their overlap analyses remain separate.

FAMILY 1

Primary curation sets

1. Paper + screen evidence
2. Apply set-specific rules
3. Resolve FlyBase identity
4. **Publish sets 01-04**

FAMILY 2

GO term-grouped sets

1. CSW FC0.5 + FC1 tables
2. DAVID · FDR ≤10%
3. Review term keywords
4. **Publish sets 05-07**

902	source observations
791	unique source FBgns
≥2	minimum pathway frequency
7	experiments

Thresholds remain separate: FC0.5 and FC1 are never collapsed into a max-frequency field.

Publication sets use explicit, reproducible rules

Sets 01–02 preserve publication evidence and resolve one biologically important identity conflict.

SET 01 · 6 GENES

Mechanistic

Publication core joined by FBgn to mechanistic screen ranks 1–6.

- Carry forward mechanism, literature status, Tx-Omics evidence, and fl.ai confidence.
- Evidence labels: consistent, correlated, or curated.
- No rank or annotation is inferred outside the source tables.

unc79 · SIFa · rumpel · AstA-R2 · Trhn · RpL23

SET 02 · 20 GENES

Homeostatic history × rebound

15

History+ / Rebound−

5

History− / Rebound+

Identity correction

Source row trh / FBgn0262139 is resolved from publication context to Trhn / FBgn0035187 (tryptophan hydroxylase). The wrong FBgn is never retained.

Screen frequency and direct-regulator evidence stay separate

Set 03 is threshold-scoped screen evidence; Set 04 comes directly from the paper.

SET 03 · 97 GENES

CSW 4+ wake-frequency union

97
FC0.5 wake frequency 4–6

7
also qualify at FC1

0
sleep genes at frequency 4

Union rule

- Take FC0.5 wake genes at frequency 4, 5, or 6.
- Add FC1 wake frequency-4 genes; all seven are already in the FC0.5 set.
- Keep FC0.5 and FC1 metadata in separate columns.

SET 04 · 4 GENES

HLH regulators

Named in Results / Table S1 after HOMER identified a CAGCTG E-box upstream of wake-induced genes.

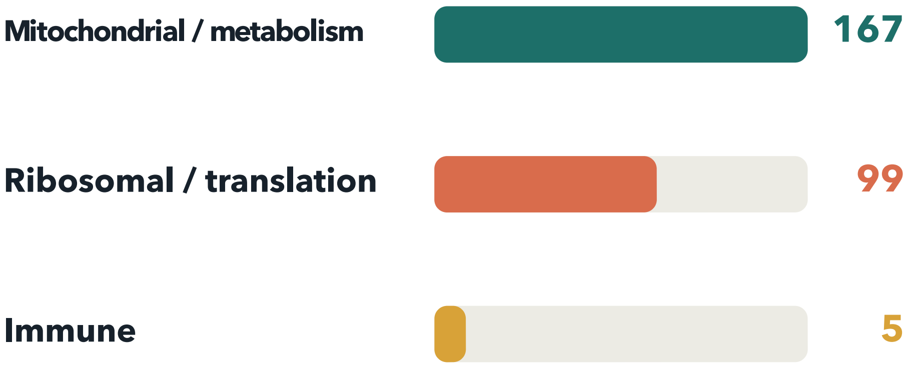
bigmax	FBgn0039509
HLH3B	FBgn0011276
E(spl)m3-HLH	FBgn0002609
E(spl)mbeta-HLH	FBgn0002733

GO-derived genes retain their CSW threshold provenance

Only hit genes from approved FDR-passing terms enter sets 05–07; FC0.5 and FC1 remain distinct.

1. **Submit four CSW masters**
2. **Keep FDR ≤10% terms**
3. **Review term keywords**
4. **Take hit FBgns**

GO term-grouped outputs



All three outputs are wake-direction genes.

Threshold provenance · 240 unique FBgns

203	FC0.5 only	Ribosomal and mitochondrial dominate
5	FC1 only	BomBc2 · BomS1 · BomS2 · IM14 · Thor
32	both thresholds	FC1 ribosomal/mitochondrial genes are also at FC0.5

Key takeaway **Only the five immune genes are FC1-exclusive.**

Identity resolution is conservative by design

Current symbols can pass directly; ambiguous synonyms require biological context or a case-supported source match.

PASS	**Trhn**	**FBgn0035187**	Current symbol · tryptophan hydroxylase neuronal · publication-resolved in Homeostatic.
REJECT	**trh**	**FBgn0262139**	Current symbol · trachealess · must not enter any generated identity field for this publication row.
REVIEW	**Trh**	**ambiguous**	Synonym of Hn, Trhn, and trh · never auto-approved without biological context.

Approval gates

- Exact current symbol
- Publication-supported correction
- Unique case-insensitive current-symbol match
- Otherwise: stop for review

Membership key

flybase_gene_id

Symbols are display labels. All overlap, deduplication, and set membership use FBgn only.

Primary-set overlap is limited to three genes

Mechanistic, Homeostatic, CSW 4+ (FC0.5), and HLH are compared only with one another.

6	20	97	4
Mechanistic	Homeostatic	CSW 4+ · FC0.5	HLH

Observed shared genes

Pairing	Count	Genes
Mechanistic × Homeostatic	2	Trhn · AstA-R2
Mechanistic × CSW 4+ (FC0.5)	1	RpL23
All other pairings	0	No shared FBgns

Within Family 1

124 unique FBgns

121 in exactly one set

3 in exactly two sets

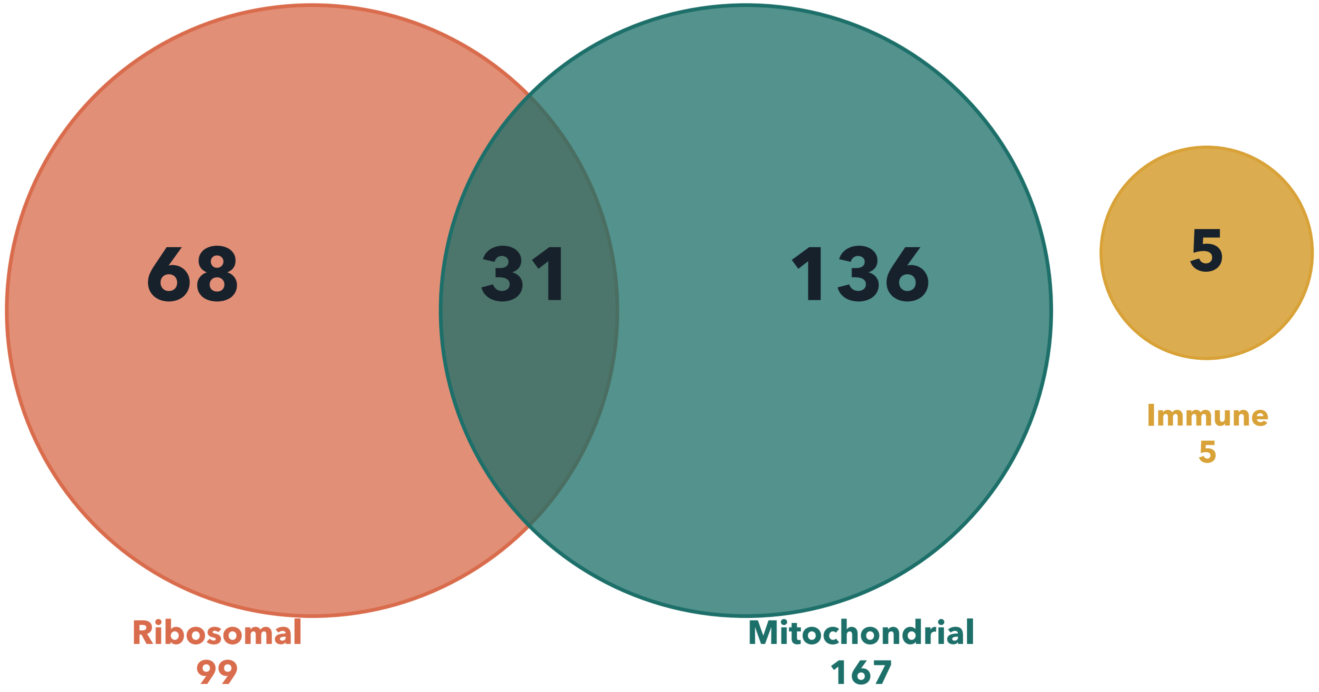
0 in three or four sets

HLH is fully disjoint from the other primary sets.

GO-derived overlap is confined to ribosomal × mitochondrial

The three pathway buckets are compared only within Family 2; threshold provenance remains a separate attribute.

GO term-group membership



240 unique FBgns in Family 2

What the overlap means

31 genes occur in both ribosomal and mitochondrial buckets

5 immune genes are fully disjoint

0 immune overlaps with either other GO bucket

The 31-gene overlap is intentional: mitochondrial translation terms match both keyword groups.

The contract is simple—and strict.

7 Exactly seven CSVs in the discovery folder

2 Required keys: ext_gene and flybase_gene_id

0 No overlap, GO, or audit CSVs beside the inputs

Seven inputs

- 01 Mechanistic
- 02 Homeostatic H×R
- 03 CSW 4+
- 04 HLH regulators
- 05 Ribosomal
- 06 Mitochondrial
- 07 Immune

RUN

```
python -m fl_ai_reagent_stocker run ./data/gene_sets/Tx-Omics_Revision/ \
  --config ./data/config/stock_split_config_priority_example.json
```

Stability rule Discovery is recursive: every extra *.csv changes the run. Keep analysis outputs elsewhere.